2025

10PEARLS

LAHORE AQI PREDICTION SYSTEM

TECHNICAL REPORT

Submitted by:

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1. SYSTEM OVERVIEW

Objectives, Targets & Tech Stacks

Goal:

- Short-horizon PM forecasting for Lahore using hourly data; predicts t+3h and t+6h for PM2.5 and PM10.

Targets:

- pm2_5_t+3h
- pm2_5_t+6h
- pm10_t+3h
- pm10_t+6h

Pipeline:

- Fully automated (data → feature engineering → train → evaluate → select best → persist → deploy)
- Random Search HPO per model.

Stack:

Data scripts (Python)	FastAPI (serving)	Streamlit (UI)	GitHub Actions (daily retrain @ 03:00 UTC / 08:00 PKT)	Hosted on Render

2. DATA & FEATURES

Dataset Collection

Data Source	Location	Window
Open-Meteo Air Quality + Weather Archive APIs	Lahore (31.5497, 74.3436)	24 months sliding, hourly

Raw columns (merged hourly):

- pm2_5
- pm10
- carbon_monoxide
- nitrogen_dioxide
- ozone
- sulphur_dioxide
- temperature_2m
- relative_humidity_2m
- wind_speed_10m
- wind_direction_10m
- time.

Post-FE dataset size

~17k+ hours

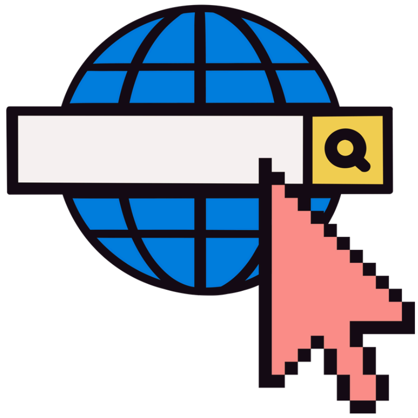
(varies per target due to lags/drops)

2.1 FEATURE GROUPS

(used for training & inference)

Category	Features	Notes
Temporal (calendar)	hour, weekday, month, is_weekend	Discrete time drivers
Temporal (cyclic enc.)	hour_sin, hour_cos, wday_sin, wday_cos, month_sin, month_cos	Preserve periodicity
Wind direction enc.	wind_dir_sin, wind_dir_cos	Angle → sin/cos
Short rolling avgs	pm2_5_avg_3h, temperature_avg_3h	Local smoothing
PM2.5 roll stats (6h)	pm2_5_mean_6h, pm2_5_std_6h, pm2_5_min_6h, pm2_5_max_6h	Recent distribution
PM2.5 roll stats (12h)	pm2_5_mean_12h, pm2_5_std_12h, pm2_5_min_12h, pm2_5_max_12h	Diurnal memory
PM10 roll stats (6h)	pm10_mean_6h, pm10_std_6h, pm10_min_6h, pm10_max_6h	Recent distribution
PM10 roll stats (12h)	pm10_mean_12h, pm10_std_12h, pm10_min_12h, pm10_max_12h	Diurnal memory
Change rates	pm2_5_change_rate, pm10_change_rate, ozone_change_rate	First-differences
Ratios & interactions	pm_ratio, temp_pm2_5, humidity_pm2_5	Composition & meteo-PM coupling

3. FEATURE GROUPS



(used for training & inference)

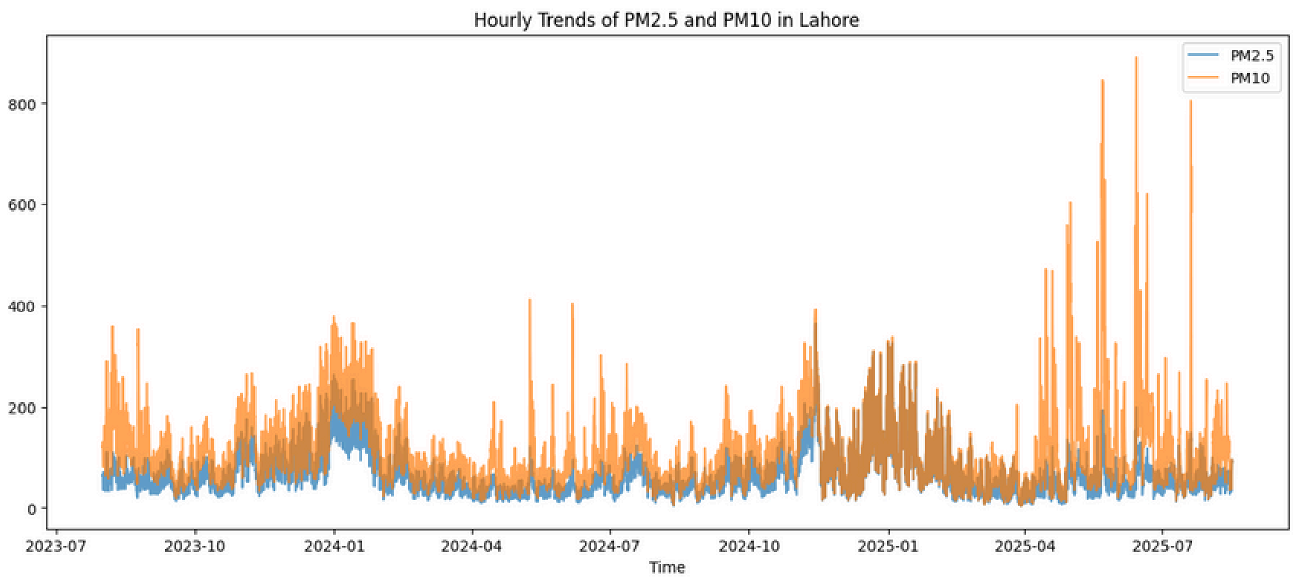
Category	Features	Notes
Lagged signals (1h,3h,24h,168h)	pm2_5_lag_1h, pm2_5_lag_3h, pm2_5_lag_24h, pm2_5_lag_168h, pm10_lag_1h, pm10_lag_3h, pm10_lag_24h, pm10_lag_168h, temperature_2m_lag_1h, temperature_2m_lag_3h, temperature_2m_lag_24h, temperature_2m_lag_168h, relative_humidity_2m_lag_1h, relative_humidity_2m_lag_3h, relative_humidity_2m_lag_24h, relative_humidity_2m_lag_168h, ozone_lag_1h, ozone_lag_3h, ozone_lag_24h, ozone_lag_168h, wind_speed_10m_lag_1h, wind_speed_10m_lag_3h, wind_speed_10m_lag_24h, wind_speed_10m_lag_168h	Multi-scale memory
Gas/Met variables	carbon_monoxide, nitrogen_dioxide, ozone, sulphur_dioxide, temperature_2m, relative_humidity_2m, wind_speed_10m	Exogenous drivers

Targets created: pm2_5_t+3h, pm2_5_t+6h, pm10_t+3h, pm10_t+6h (shifted forward by 3/6 hours).

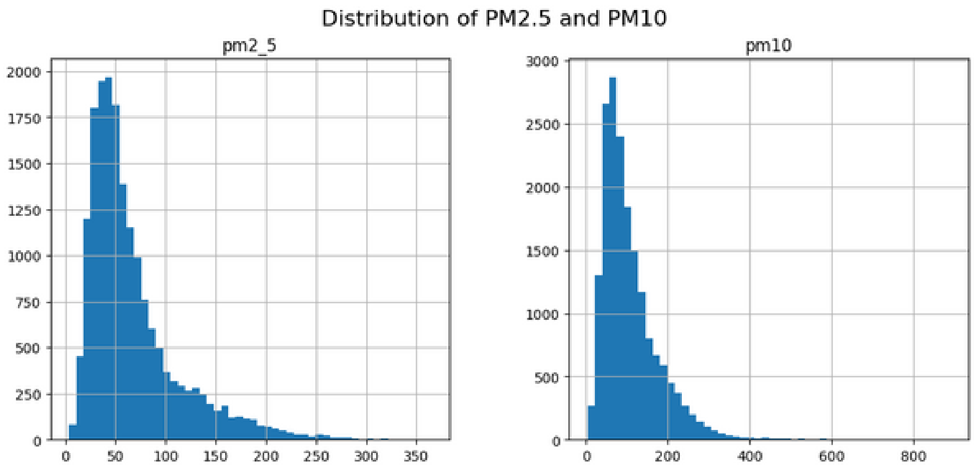
3.1. LAHORE HOURLY DATASET - EDA

Performing analysis on dataset

PM2.5 and PM10 Trends Over Time:



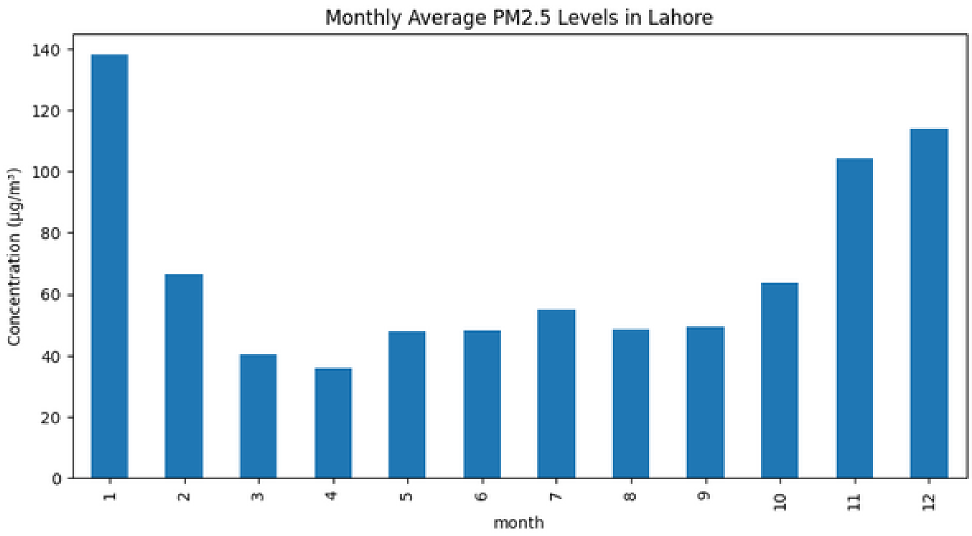
PM2.5 and PM10 Trends Over Time:



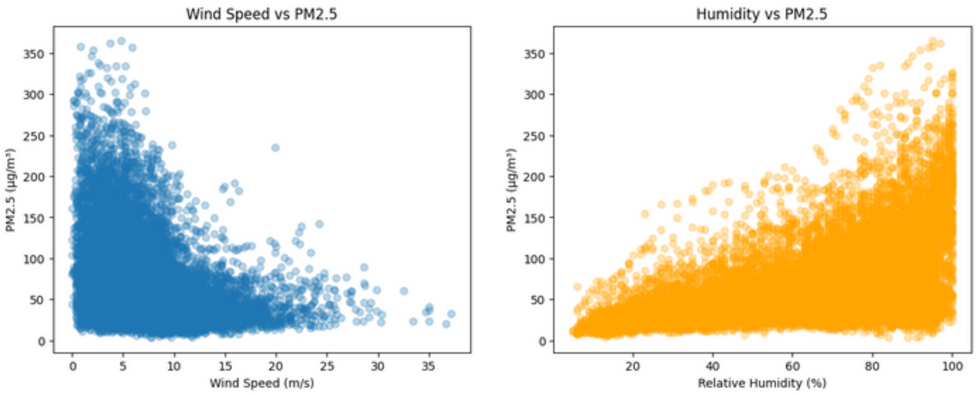
3.1. LAHORE HOURLY DATASET - EDA

Performing analysis on dataset

Seasonal Pattern (Monthly Averages):



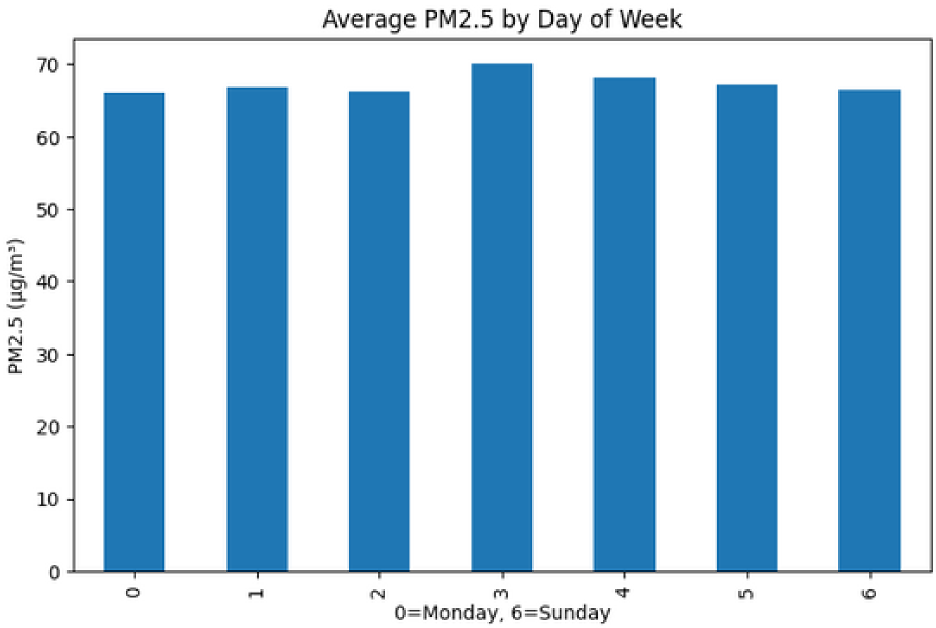
Impact of Weather (Scatter Plots):



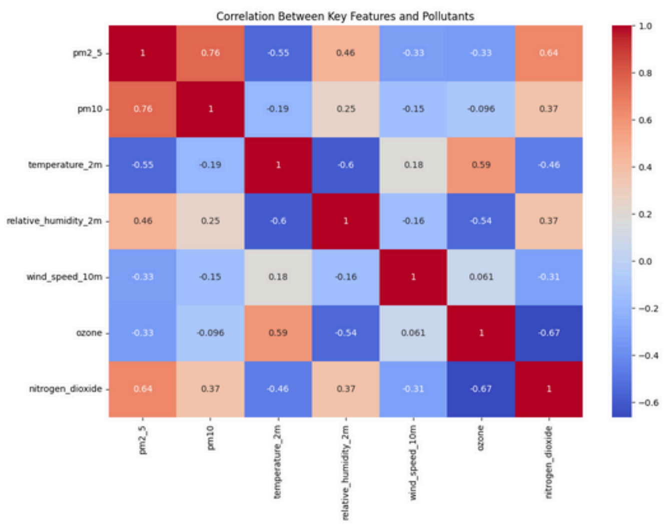
3.1.
LAHORE HOURLY DATASET - EDA

Performing analysis on dataset

Weekly Cycle:



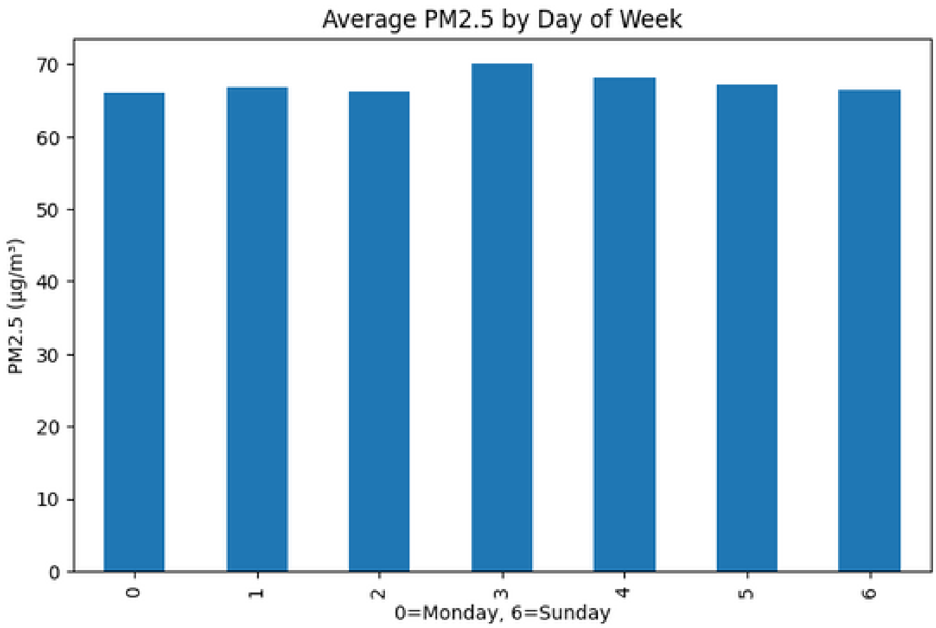
Correlation Heatmap:



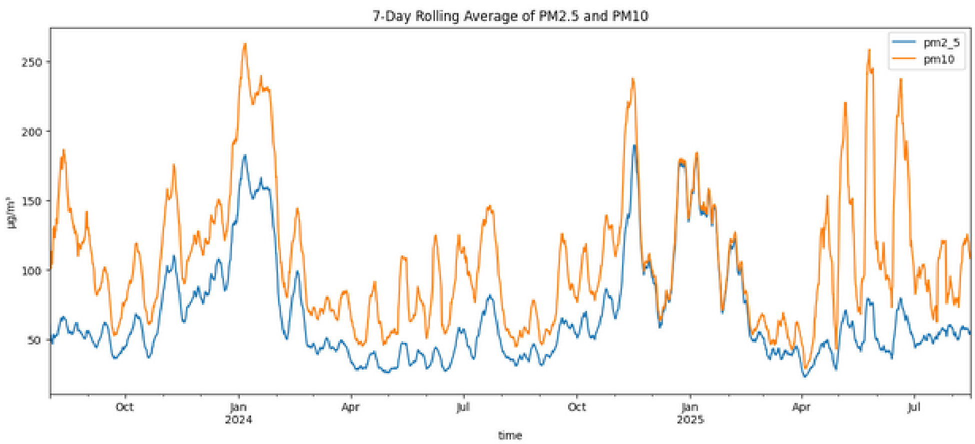
3.1. LAHORE HOURLY DATASET - EDA

Performing analysis on dataset

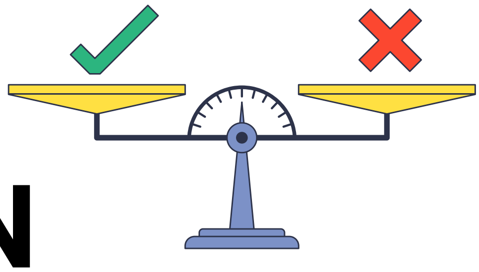
- **Rolling Averages (Smoothing):**



- **AQI Category Breakdown:**



4. RESULTS COMPARISON



(All metrics from the detailed training evaluation you provided.)

Winner

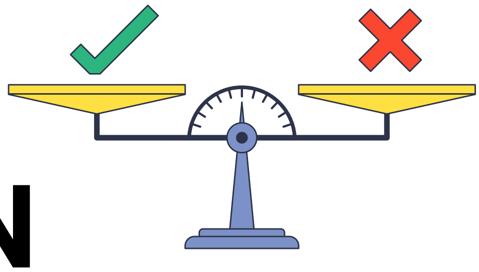
0.9456 CV-Score

(Catboost Regressor)

4.1 PM2.5 — t+3h:

model	cv_score	r2	mse	rmse	mae
catboost_regressor	**0.9456**	**0.9456**	**122.4120**	**11.0640**	**7.7529**
lightgbm_regressor	0.9455	0.9455	122.6585	11.0751	7.7275
xgboost_regressor	0.9441	0.9441	125.8739	11.2194	7.6500
gradient_boosting_regressor	0.9094	0.9380	139.6360	11.8168	8.2425
linear_regression	0.8909	0.8909	245.6274	15.6725	10.8842
extra_trees_regressor	0.8838	0.9139	193.7283	13.9186	9.7715
random_forest_regressor	0.8808	0.9150	191.3496	13.8329	9.6329
lasso_regression	0.8763	0.8905	246.5981	15.7034	10.8975
elastic_net_regression	0.8762	0.8906	246.2881	15.6936	10.8960
ridge_regression	0.8761	0.8909	245.6376	15.6728	10.8838
decision_tree_regressor	0.8090	0.8692	294.4222	17.1587	11.5921

4. RESULTS COMPARISON



(All metrics from the detailed training evaluation you provided.)

Winner

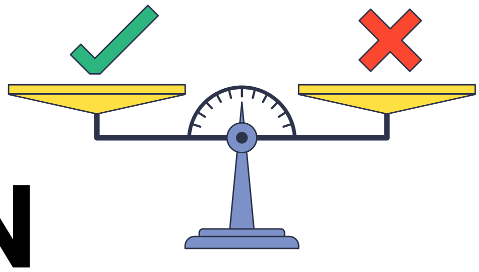
0.9274 CV-Score

(LightGBM Regressor)

4.2 PM2.5 — t+6h:

model	cv_score	r2	mse	rmse	mae
lightgbm_regressor	0.9274	**0.9274**	**164.8463**	**12.8392**	**9.2498**
xgboost_regressor	0.9238	0.9238	172.8806	13.1484	9.2889
catboost_regressor	0.9165	0.9165	189.5707	13.7685	10.0121
gradient_boosting_regressor	0.8372	0.9024	221.4889	14.8825	10.8553
extra_trees_regressor	0.8110	0.8797	272.9967	16.5226	12.2832
random_forest_regressor	0.8006	0.8851	260.8203	16.1499	11.8253
linear_regression	0.7676	0.7676	527.5615	22.9687	16.5439
lasso_regression	0.7437	0.7671	528.6261	22.9919	16.5590
ridge_regression	0.7433	0.7672	528.3899	22.9867	16.5633
elastic_net_regression	0.7433	0.7673	528.3490	22.9858	16.5666
decision_tree_regressor	0.6692	0.8065	439.2695	20.9588	14.2615

4. RESULTS COMPARISON



(All metrics from the detailed training evaluation you provided.)

Winner

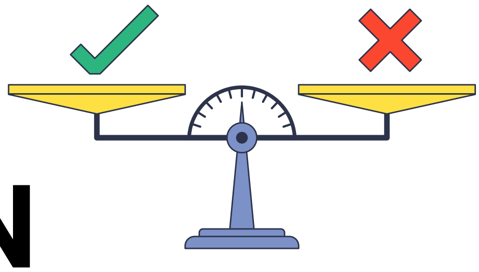
0.9039 CV-Score

(LightGBM Regressor)

4.3 PM10 — t+3h:

model	cv_score	r2	mse	rmse	mae
lightgbm_regressor	0.9039	0.9039	538.3272	23.2019	14.6270
xgboost_regressor	0.8969	0.8969	577.9086	24.0397	14.9240
catboost_regressor	0.8953	0.8953	586.7115	24.2221	15.1308
gradient_boosting_regressor	0.8364	0.8844	647.4652	25.4453	15.8506
extra_trees_regressor	0.8302	0.8636	764.0949	27.6423	18.0333
random_forest_regressor	0.8166	0.8808	667.7050	25.8400	15.8612
linear_regression	0.8029	0.8029	1104.6271	33.2359	19.4173
elastic_net_regression	0.8000	0.7969	1137.9900	33.7341	19.5872
ridge_regression	0.7997	0.8008	1116.3062	33.4112	19.4505
lasso_regression	0.7987	0.7966	1139.4470	33.7557	19.5065
decision_tree_regressor	0.7338	0.7966	1139.5856	33.7577	21.9910

4. RESULTS COMPARISON



(All metrics from the detailed training evaluation you provided.)

4.4 PM10 — t+6h:

Winner

0.8397 CV-Score

(XGBoost Regressor)

model	cv_score	r2	mse	rmse	mae
xgboost_regressor	**0.8397**	**0.8397**	**888.5228**	**29.8081**	**18.2753**
lightgbm_regressor	0.8303	0.8303	940.8947	30.6740	18.8686
catboost_regressor	0.8231	0.8231	980.7221	31.3165	19.9020
gradient_boosting_regressor	0.6817	0.7940	1142.2937	33.7978	21.2701
extra_trees_regressor	0.6651	0.8165	1017.4587	31.8976	19.4762
random_forest_regressor	0.6521	0.7903	1162.4203	34.0943	20.3163
linear_regression	0.6001	0.6001	2217.1929	47.0871	28.9398
elastic_net_regression	0.5887	0.5957	2241.2630	47.3420	29.0792
ridge_regression	0.5886	0.5989	2223.5969	47.1550	28.9895
lasso_regression	0.5857	0.5944	2248.8469	47.4220	29.1124
decision_tree_regressor	0.4526	0.5993	2221.2376	47.1300	30.6313

4.5
LATEST TRAINED RUN (2025-08-15)

Best Selected Models for each Target

FInal Selected Models:

Target	Model	CV score	RMSE	MSE	R^2
pm2_5_t+3h	**CatBoost**	0.942249	11.50574	132.38205	0.942249
pm2_5_t+6h	**LightGBM**	0.924752	12.99878	168.96836	0.924752
pm10_t+3h	**LightGBM**	0.904192	23.06641	532.05931	0.904192
pm10_t+6h	**XGBoost**	0.841179	28.89465	834.90086	0.841179

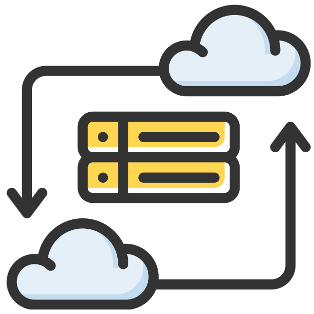
Artifacts:

- models/<target>_best.joblib, models/metrics.json

SHAP:

- models/shap/<target>_shap.csv

5. DEPLOYMENT & OPS



Deployment environment details

Hosting: Render (API + Streamlit).

Scheduler: GitHub Actions, daily 03:00 UTC (08:00 PKT).

Job steps: collect data → build features → train & evaluate (Random Search) → write metrics & SHAP → persist best → reload API.

Observability: /health (status + which models loaded), logs from Render & Actions.

Config: CORS_ORIGINS (FastAPI), LAT/LON, MONTHS_BACK for data script.

6. API REFERENCE & USAGE

Base URL: Render deployment (FastAPI). CORS enabled per CORS_ORIGINS.

6.1 Endpoints (summary)

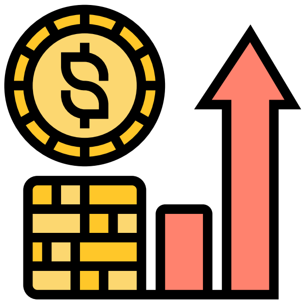
Method	Path	Purpose	Notes
GET	/	Liveness ping	{ ok: true, message: "AQI API ready" }
GET	/reload	Reload features/models/raw into memory	Returns which models are missing
GET	/health	Status of loaded models & features	status: ok/degraded, loaded_models, features_count, raw_loaded
GET	/features	Feature list used for inference	From features_used.json
GET	/models	Latest training metrics	Reads models/metrics.json
GET	/timeseries?hours=168	Recent raw series snapshot	Returns arrays for time, pm2_5, pm10, gases
GET	/explanations?target=pm2_5_t+3h&top_k=20	SHAP summary for a target	Requires precomputed SHAP CSV
GET	/predict?horizon=3	Forecast endpoint	horizon ∈ {3,6}; returns PM2.5 & PM10 for that horizon + EPA category

7.
FINAL REMARKS (PER TARGET)

Summarize the results per target

PM2.5 t+3h (CatBoost)	PM2.5 t+6h (LightGBM)	PM10 t+3h (LightGBM)	PM10 t+6h (XGBoost)
Strongest overall; high CV (≈0.94–0.95). Great short-horizon sensitivity to recent lags and rolling stats.	Stable medium horizon; slight accuracy dip vs. t+3h as expected.	Competitive despite noisier PM10 dynamics; benefits from lagged PM10 and met variables.	Hardest target; XGBoost leads; consider stacked LGBM+XGB (weights) for small uplift (supported by API bundle logic).

8.
FUTURE ENHANCEMENT



Next Tweaks (roadmap)

Future upgrades

- Add **holiday/event flags** and **precipitation features** (if available) to capture spikes.
- Enable **probabilistic forecasts (quantile regression)** for uncertainty bands in UI.
- Add basic **data quality checks (gap detection, sensor sanity)** in CI.
- Persist **feature importance charts** to surface key drivers per target in Streamlit.

APPENDIX – REPRO & FILES

Artifcats and document links

- scripts/data_collect_update.py (collection + FE + export per target)
- data/raw_lahore_hourly.csv, data/lahore_features_no_targets.csv
- datasets_per_target/<target>.csv
- models/<target>_best.joblib, models/features_used.json, models/metrics.json, models/shap/<target>_shap.csv

GitHub Repository

https://github.com/ByteCraftByLaiba/aqi-prediction-system.git

Deployed Project Link

Render

https://aqi-ui.onrender.com/